

EfficientLLaVA: Generalizable Auto-Pruning for Large Vision-language Models

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Abstract

While multimodal large language models demonstrate strong performance in complex reasoning tasks, they pose significant challenges related to model complexity during deployment, especially for resource-limited devices. In this paper, we propose an automatic pruning method for large vision-language models to enhance the efficiency of multimodal reasoning. Conventional methods rely on the training data of the original model to select the proper pruning ratio for different network components. However, these methods are impractical for large vision-language models due to the unaffordable search costs caused by web-scale training corpus. In contrast, our approach only leverages a small number of samples to search for the desired pruning policy by maximizing its generalization ability on unknown training data while maintaining the model accuracy, which enables the achievement of an optimal trade-off between accuracy and efficiency for large visual language models. Specifically, we formulate the generalization gap of the pruning strategy using the structural risk minimization principle. Based on both task performance and generalization capability, we iteratively search for the optimal pruning policy within a given search space and optimize the vision projector to evolve the search space with higher upper bound of performance. We conduct extensive experiments on the ScienceQA, Vizwiz, MM-vet, and LLaVA-Bench datasets for the task of visual question answering. Using only 64 samples for pruning policy search, EfficientLLaVA achieves an accuracy of 83.05% on ScienceQA, along with a $\times 1.8$ speedup compared to the dense LLaVA-v1.5-7B model. The code is available at: <https://github.com/liangynTHU/EfficientLLaVA>

1. Introduction

Multimodal large language models have been widely adopted in complex reasoning tasks such as visual question answering [10, 34], embodied task planning [12, 28] and dialogue

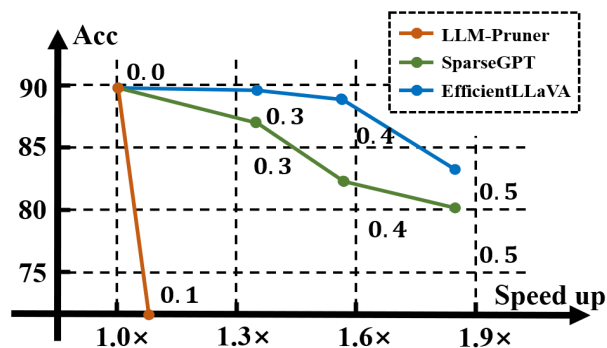


Figure 1. The trade-off between accuracy and inference speed on ScienceQA with EfficientLLaVA.

systems [16]. Although they achieve excellent performance in high-level tasks with rich commonsense, the inference stage of large vision-language models (LVLMs) in deployment requires large memory footprint and long latency because of the tremendous neurons. In order to deploy the powerful LVLMs on mobile devices such as cellphones, navigation robots, and autonomous vehicles with a strict computational resource limit, we are required to reduce the model complexity without obvious degradation of performance.

Network pruning aims to remove unimportant components including layers [38], channels [25], tokens [41, 42], and neurons [44] to reduce the storage and computational complexity without harming the task performance, which have been widely studied for computer vision [4, 13], natural language processing [25, 33] and system control [36]. The components importance can be defined as the l_1 and l_2 magnitude of weights and activations [35], and the Jacobian and Hessian matrix regarding the loss functions [30]. Since different components usually have various sensitivity to pruning in downstream tasks, automatic pruning assigns the optimal pruning ratio to each component in order to achieve higher accuracy-efficiency trade-off. In conventional methods, the training data of the original model is leveraged to evaluate the sampled pruning policy based on the accuracy and the model complexity, which provides feedback to

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the search algorithm to select better candidates. However, LVLMs require extensive training corpus, and the search cost associated with existing automatic pruning methods is prohibitively high.

In this paper, we present EfficientLLaVA that automatically prunes LVLMs to achieve low model complexity and unaffected task performance with an acceptable search cost. Unlike existing automatic pruning methods that leverage the large-scale training data of the original models for pruning policy search, our method only uses a few samples to assign the optimal pruning ratio for weight matrix in each layer. Since we maximize the generalization ability of the pruning policy to the unknown web-scale training corpus, the acquired solution achieves a satisfying trade-off between accuracy and efficiency in a wide variety of downstream tasks. More specifically, we first formulate the generalization gap of the pruning policy between the proxy samples for policy search and the unknown training set via structural risk minimization principle, which can be evaluated by the Frobenius norm of the weight matrix in the pretrained LLMs. We then evolve the pruning policy candidates with the goal of achieving higher task performance and generalization ability, where the weights of MLP and attention are pruned based with the selected pruning ratio. The vision projector is optimized to enhance the upper bound of accuracy and the generalization ability for all policies in the pruning space, which is estimated via the Euclidean distance between the sampled policy and neighbor candidates. The policy candidate selection and the pruning space evolution are implemented iteratively to acquire satisfying accuracy-efficiency trade-off in the extremely large pruning space with acceptable cost. Fig. 1 demonstrates the trade-off between accuracy and inference latency on ScienceQA for uniform pruning methods SparseGPT [7] and our EfficientLLaVA. We conducted extensive experiments in a wide variety of multimodal reasoning tasks including short-answer, option-only for multiple-choice and natural QA, and the results shows that EfficientLLaVA can achieve 83.05% accuracy on ScienceQA and $\times 1.8$ speedup compared to the dense LLaVA-v1.5-7B model. Our contributions can be summarized as follows:

- We introduce EfficientLLaVA, a novel pruning method for large vision-language models that effectively reduces model complexity while preserving task performance with minimal data usage.
- Our approach leverages the structural risk minimization principle to maximize the generalization ability of pruning policies, enabling efficient adaptation to unknown training data.
- Extensive experiments demonstrate that EfficientLLaVA achieves a significant speed-up in inference while maintaining competitive accuracy across various multimodal reasoning tasks.

2. Related Work

Large Vision-language Models: Large Vision-language Models (LVLMs) fuse visual and linguistic modals via bridging the gap between different modalities that achieved outstanding performance, such as in-context predictions [22, 32], multi-image and chain-of-thought reasoning [6, 40]. To utilize a learnable connector module projecting visual information into language modal space, token-level fusion and feature-level fusion are presented for encoding multimodal information to LLMs. For token-level fusion, LLaVA [20] series utilized linear MLPs to align the image tokens with language tokens, finetuning LVLMs with visual instruction samples to enhance zero-shot capabilities. MM1 [27] compared the significance of various architectural aspects and found that the number of visual tokens and input resolution were more important than the projector design. For feature-level fusion, Flamingo [1] inserted extra cross-attention layers to connect pre-trained vision-only and language-only models, significantly boosting learning efficiency. CogVLM [37] integrated a visual expert module into every Transformer layer to fuse vision and language features. Existing methods usually focus on the performance of LVLMs, ignoring real-world deployment particularly in environments with limited resources. Despite light-weight LVLMs such as MobileVLM [5], TinyLLaVA [45], and quantized methods such as GPTQ [8], AWQ [17] have reduced the size of LLMs, the cost still exceeds the resource constraint of mobile devices and they fail to consider the intermodal interaction. Differently, our method focuses on pruning the pretrained LVLMs to find the optimal trade-off between accuracy and efficiency with few-shot proxy samples.

Automatic Pruning: Automatic pruning [18, 24] removes unnecessary connections, neurons, and layers to reduce the model size, while the model accuracy remains high. The early work of automatic pruning AMC [11] utilized reinforcement learning to effectively explore the design space automatically, thereby enhancing the quality of model compression. To achieve fine-grained pruning across samples, dynamic pruning is proposed to predict an instance-wise pruning strategy for each input sample. AutoPrune [39] pruned the network by optimizing a set of trainable parameters instead of original weights. DynamicVit [31] progressively and adaptively removed unnecessary tokens based on the input due to the sparse attention. AutoCompress [21] introduces an automatic structured pruning framework that improves model compression and inference speed using ADMM-based pruning and heuristic search, achieving high pruning rates. When pruning large pre-trained neural networks, the training set for the original model cannot be acquired. To address this, post-training pruning [14, 15] derives the sensitivity of each component to pruning by first-order and second-based gradient information. Dynamic Context Pruning [2] utilized a learnable mechanism that selects uninformative tokens to be

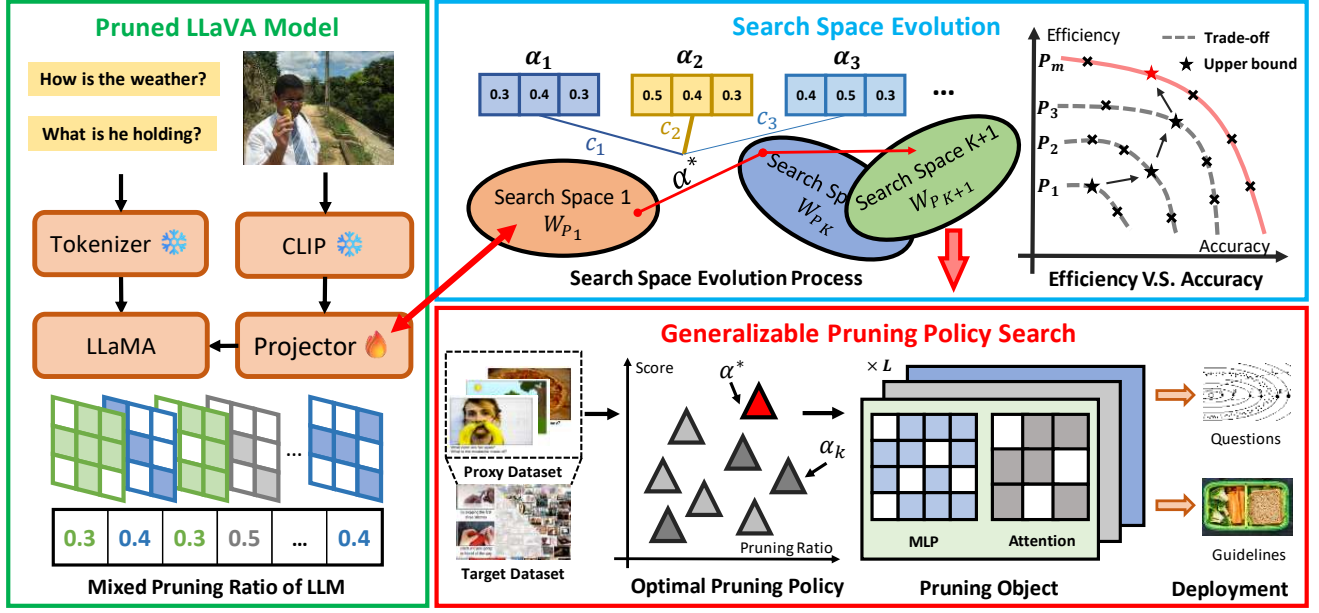


Figure 2. The overall pipeline of EfficientLLaVA. In each iteration, we first search for the optimal pruning policy for matrix in each LLaMA layer, where evolutionary algorithms are employed with the fitness function containing model accuracy and generalization ability. Then we evolve the search space by optimizing the projector weight so that the upper bound of accuracy and generalization ability for all policies can be improved.

removed from the context at various stages of the generation process. However, traditional automatic pruning methods are not satisfied for LVLMs due to ignoring the generalization gap between training data and few-shot proxy samples. Different from the previous works, our method evolve the search space to find the optimal pruning policy maximizing the generalization ability.

3. Method

In this section, as illustrated in Fig. 2, we first introduce the preliminary of automatic pruning for LVLMs. We then formulate the generalization gap between the few-shot proxy samples for pruning policy search and the overall data distribution, and search the pruning policy with generalizability maximization in given search space. Finally, we evolve the search space to improve the upper bound of accuracy and generalizability ability for all policies. We ensure that the search space remains flexible and evolves dynamically with each iteration, improving the overall pruning performance.

3.1. Automatic Pruning for LVLMs

Model pruning aims to remove unnecessary or redundant elements to reduce the size of neural network and computational complexity, while minimizing the impact on performance. Automatic pruning assigns the optimal pruning ratio to different components to achieve the optimal trade-off between accuracy and efficiency. However, conventional automatic

pruning methods [2, 9] search the pruning policy on large-scale training data, which is impractical for LVLMs due to the prohibitive training cost and the lack of the training corpus. The goal of automatic model pruning is to minimize loss objective with the sparsity constraint:

$$\min_{\alpha} L_{train}(\hat{\mathbf{M}}(\alpha), \alpha) \\ s.t. \quad \|\alpha\|_1 = C_0, \quad \hat{\mathbf{M}}(\alpha) = \arg \min L_{train}(\mathbf{M}, \alpha) \quad (1)$$

where L_{train} represents the loss in the large-scale training set for LVLMs and $\mathbf{M}$ is the mask of weight. α refers to the pruning policy, which determines how much of each layer or component to prune. The pruning policy α denotes sparsity ratio for weight matrix in each layer, and $\|\cdot\|_1$ means the L_1 norm. This ensures that a certain proportion of the model’s parameters are removed (pruned) while minimizing the impact on accuracy. C_0 stands for the overall sparsity limit for the entire LVLM. The optimal pruning masks for weights $\hat{\mathbf{M}}(\alpha)$ are acquired by minimizing the loss on the training data. Due to the limited proxy samples during pruning policy search of pre-trained LVLMs, the acquired policy usually overfits the proxy data without generalization to the entire dataset. Therefore, we should also consider the generalization gap between the provided proxy samples and the real training data distribution for the pruning policy, and we can obtain pruning policies with satisfying accuracy-efficiency trade-off in diverse downstream tasks.

3.2. Generalizable Pruning Policy Search for LVLMs

Evaluating the true network objective for searched pruning policies is very challenging because of the limited samples compared with the entire training data. Therefore, we maximize the generalization ability of the pruning policy via the structural risk minimization (SRM) principle, which can be bounded by empirical risk and unseen data distribution. Structural Risk Minimization (SRM) is a principle of statistical learning theory that seeks to balance model complexity with training data performance, minimizing the risk on both known and unseen data. The optimal trade-off between complexity and model performance ensures that our approach can be applied effectively in various real-world applications. For a dataset containing N samples in pruning policy search, the structural risk is written in the following with the probability at least $1 - \epsilon$:

$$\mathbb{E}_r[L(\mathbf{M}, \alpha)] \leq \mathbb{E}_e[L(\mathbf{M}, \alpha)] + 2\mathcal{R}(\mathcal{F}) + \sqrt{\frac{\ln 4/\epsilon}{N}} \quad (2)$$

where $\mathbb{E}_r$ and $\mathbb{E}_e$ are the expectation over the unknown real data distribution and the empirical expectation over limited proxy samples for policy search. $\mathcal{R}(\mathcal{F})$ means the Rademacher complexity in the objective function class $\mathcal{F}$. A lower Rademacher complexity indicates better generalization on unseen data. Since the empirical risk represents the performance of the pruning policy on the limited provided samples, we minimize the Rademacher complexity to reduce the gap between the empirical risk and the real structural risk. The Rademacher complexity means the upper bound of the weighted sum of objectives across samples for all possible objectives. Following [29], we efficiently evaluate the Rademacher complexity via the matrix norm for all layers in the model:

$$\mathcal{R}(\mathcal{F}) = \frac{1}{N} \cdot \mathbb{E}_\sigma \left[\sup_{L \in \mathcal{F}} \sum_{i=1}^N \sigma_i L_{x_i}(\mathbf{M}, \alpha) \right] \propto \eta \prod_{i=1}^K \mathcal{A}_i \quad (3)$$

where L_{x_i} represents the loss function of the i_{th} sample x_i for policy search, and $\{\sigma_i\}$ are independent random variables drawn from the Rademacher distribution. $\mathcal{A}_i$ stands for the Frobenious norm of the parameter matrix in the i_{th} layer, and η can be regarded as a constant related to the layer number and the sample size. The low weight norm indicates that the model output is weakly correlated with the input, which leads to high generalization ability for different data distributions. Despite selecting components with satisfying accuracy-efficiency trade-offs on the limited proxy samples, we also encourage the remaining components to have low weight norm. Therefore, the acquired pruning policy can be adapted to downstream tasks with unseen diverse data distribution.

Since the LLaMA model in multimodal LLMs contributes significantly to the overall model complexity, we prune the weight matrix for each MLP in the LLaMA model. The reason for excluding the visual encoder and projector from pruning is that pruning these components minimally impacts computation, but can cause significant performance degradation. The k_{th} element of α represents the pruning ratio of the k_{th} MLP layer. The fitness function J for generalizable search can be written as follows:

$$\max_{\alpha} J(\alpha) = \mathbb{E}_e[L(\mathbf{M}, \alpha)] + \eta \prod_{i=1}^K \mathcal{A}_i(\mathbf{M}_i, \alpha_i) \quad (4)$$

where $\mathcal{A}_i(\mathbf{M}_i, \alpha_i)$ means the matrix norm of the i_{th} layer for the pruning ratio α_i and the pruning mask $\mathbf{M}_i$. Given the pruning ratio for each layer, we acquire the pruning mask by OBS algorithm [7] removing weights that have the lowest influence on the output of layer. The OBS (Optimal Brain Surgeon) algorithm optimally removes weights while minimizing the increase in loss, thus preserving model performance. We employ evolutionary algorithms to search for the optimal pruning policy.

3.3. Search Space Evolution of Pruning Policies for LVLMs

LVLMs align the visual tokens with the language tokens via the projector layer, so that the knowledge learned in the pretrained CLIP visual encoder and the LLaMA model can be fully leveraged for multimodal reasoning. Therefore, we can optimize the weights of the projector, where the possible highest accuracy and generalizability can be acquired for pruning policies given the resource budget. In this context, the projector plays a crucial role in fusing visual and language modalities, making it a key target for optimization in ensuring high model performance even after pruning. The projector weights are regarded as the search space of the pruning policy search in LVLMs, and optimizing the projector weights is equivalent to the search space evolution. The objective for the evolution is as follows:

$$\max_{\mathbf{W}_p} \sup_{\alpha} J(\mathbf{W}_p, \alpha) = \sup_{\alpha} \left[\mathbb{E}_e + \eta \prod_{i=1}^L \mathcal{A}_i(\mathbf{M}, \alpha) \right] \quad (5)$$

where $J(\mathbf{W}_p, \alpha)$ means the fitness function with respect to the projector weight $\mathbf{W}_p$ and the pruning policy α , and $L(\mathbf{W}_p, \mathbf{M}_w, \alpha)$ means the loss function of the model regarding the projector weight despite of the pruning mask and the pruning policy. The optimization goal here is to find the best combination of projector weights and pruning ratios to maximize performance. This step ensures that the search space continually adapts and improves, rather than being static. By maximizing the upper bound of the fitness function with search space evolution, pruning policies with higher performance and generalizability appear in the evolved search

space. However, estimating the upper bound of the fitness function over all pruning policies is infeasible as enumerating the countless selections is computationally prohibited. We relax the original problem as the following one with the weighted fitness function $\hat{J}$:

$$\max_{\mathbf{W}_p} \hat{J}(\mathbf{W}_p, \alpha) = \mathbb{E}_e[c_m L_m(\mathbf{W}_p, \mathbf{M}, \alpha)] \quad (6)$$

where c_m and L_m represent the importance weight and the loss function for the m_{th} candidate pruning policy in the current search space. The candidate that is closer to the upper bound should be considered with higher importance in the search space evolution, and the corresponding importance weight should be larger. Meanwhile, the matrix norm is independent of the projector weights, and it is omitted in the optimization objective of the projector weights. The importance weight is estimated via the following rule:

$$c_m = \frac{\exp(-\sum_{\alpha \in \mathcal{N}_m} \|\alpha_m - \alpha\|^2)}{\sum_k \exp(-\sum_{\alpha \in \mathcal{N}_k} \|\alpha_k - \alpha\|^2)} \quad (7)$$

where $\mathcal{N}_m$ means the neighborhood of the m_{th} candidate pruning policy defined by Euclidean distance. The normalized importance weight c_m assigns higher values to candidates with closer neighbors, implying that candidates in dense regions of the search space (close to other candidates) are considered more important. Different candidates tend to converge towards the optimal pruning policy, and a policy with neighbors at shorter distances suggests a higher likelihood of being optimal, as it is closer to the optimal pruning solution. This weight calculation ensures that pruning policies with stronger potential for optimality are prioritized during the search process, improving the overall quality of the evolved search space.

The overall pipeline for automatic pruning of LVLMs is demonstrated in Algorithm 1. For a pre-trained multimodal LLM, we first search for the optimal pruning policy with the limited proxy samples given the search space represented by the projector weights. Then we remove the unimportant weights for each layer with the searched pruning ratio via the OBS pruning method. Finally, we optimize the projector weight for search space evolution. The above three stages are iteratively implemented in each round until reaching the search cost constraint.

4. Experiments

In this section, we conduct extensive experiments for LLaVA architectures on multi-modal question answering datasets. Firstly, we introduce the implementation details of our EfficientLLaVA and the dataset information. Then we conduct ablation study to evaluate the effectiveness of the generalizable pruning policy search and the search space evolution of pruning policies. Finally, we compare the performance re-

Algorithm 1: Search Space Evolution of Generalizable Pruning Policy

Input : Initial search space $\mathbf{W}_p^0$, Pruning ratio constraint C , max evolution step τ , the number of pruning policy sampling n , the size of evolution group

Output : The optimal pruning ratio α , search space $\mathbf{W}_p$

- 1 $G_0 :=$ Randomly sample n pruning policy $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ with the constraints $\mathcal{C}$
 - 2 **while** evolution step $t \leq \tau$ **do**
 - 3 Prune networks base on the OBS pruning method and pruning policy G_t in the t_{th} round
 - 4 Obtain top- k candidates with fitness function (4)
 - 5 Generate G_{t+1} according to evolutionary algorithm
 - 6 Predict the optimal pruning policy via Euclidean distance in (7)
 - 7 Evolve the search space $\mathbf{W}_p^t$ base on (6)
 - 8 **end**
-

garding accuracy and efficiency with existing pruning methods.

4.1. Setups

Dataset: We conduct the experiments on ScienceQA [23], Vizwiz [3], MMVet [43] and LLaVA-Bench [20]. ScienceQA is the first large-scale multimodal dataset that annotates lectures and explanations for the answers containing 21k multiple-choice science questions. Moreover, ScienceQA dataset encompasses vision-language pairs that can be categorized into subjects such as natural science (NAT), social science (SOC), and language science (LAN). Additionally, another classification based on context modality includes text context (TXT), image context (IMG), and no context (NO). ScienceQA dataset stands out for having the largest collection of images, covering all 12 grades, containing the longest questions, and featuring the most diverse input sources. Vizwiz arose from a natural setting for visual question answering with blind individuals, together with 10 crowdsourced answers per visual question, contributing images, and spoken questions, which contains 20k image-question pairs. MMVet leverages the amalgamation of various core visual-linguistic competencies to address complex challenges, which defines 16 emergent tasks of interest integrated from the six defined core VL capabilities. LLaVA-Bench is a benchmark designed to test how well multimodal AI models can handle real-world visual tasks by engaging in open-ended conversations based on diverse images. These datasets present distinct challenges, from the complexity of real-world queries to specialized reasoning tasks, ensuring

Table 1. Comparison of average pruning ratio and the accuracy evaluated on ScienceQA across different pruning method.

Criteria	Avg ratio	Accuracy
Uniform	0.50	80.45
Auto w/o Gen	0.49	81.34
EfficientLLaVA	0.50	83.05

that EfficientLLaVA’s performance is tested under a variety of realistic conditions.

Implementation Details We evaluate our automatic pruning method for pretrained LVLMs including LLaVA-SQA-7B [20], which is finetuned on ScienceQA dataset and LLaVA-v1.5-7B [19], which achieves SoTA on a broad range of 11 tasks. The average pruning ratio p_0 for the entire model is set to 0.3, 0.4 and 0.5. To ensure the average pruning ratio of all candidate pruned networks remains consistent, the choices of pruning ratio for each layer is selected from $[p_0 - 0.1 : 0.05 : p_0 + 0.1]$. This ensures the pruning process adapts to the structure of each layer without significantly affecting the model’s integrity. We also set the upper and lower bounds for the generated candidate pruning ratios as $p_0 + 0.01$ and $p_0 - 0.01$. This range is carefully chosen to allow sufficient flexibility for pruning each layer while maintaining overall consistency with the target pruning ratio. For proxy data, we use 64 segments with 256 tokens for each segment which are randomly sampled from the PTB dataset [26], which is versatile and has been used in various tasks, and covers a wide range of English language constructs, making it a valuable resource for understanding and modeling the syntactic and semantic aspects of the language. It embodies generic data extracted from Internet sources, which indicates the zero-shot settings of our experiments because no task-specific data is provided. We evolve 16 policy candidates in each round and the value of the hyperparameter η in LLaVA-SQA-7B pruning is set to 0.5. This number of policy candidates ensures sufficient exploration of the pruning space without incurring excessive computational overhead. Given the searched pruning ratio, we use the same pruning methods as in SparseGPT [7] for model pruning to fully leverage the potential in network pruning with high degrees of freedom. We also sparsify Transformer layers sequentially and only prunes the matrix of attention weights and MLP weights in the pre-trained LLaMA model. During search space evolution, we select top-5 nearest neighbors to compute the importance weight in (7). The number of rounds containing policy search, weight pruning, and search space evolution is 10.

4.2. Ablation Study

Effects of generalizable automatic pruning: Automatic pruning determines optimal pruning ratios for various components to achieve the ideal trade-off between accuracy and

Table 2. The effects of search space evolution regarding the average pruning ratio and the accuracy.

Technique	Avg ratio	Accuracy
No Evo	0.50	81.07
Evo w/o UB	0.51	82.85
EfficientLLaVA	0.50	83.05

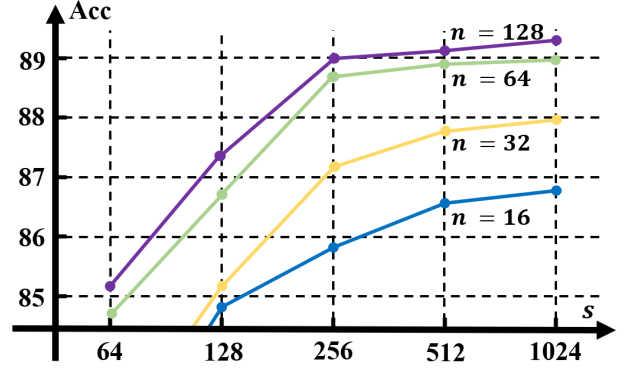


Figure 3. The performance variation with the number of sampled data n and sequence length s in pruning space evolution tested on ScienceQA.

efficiency. In order to minimize the generalization gap between the provided samples and training data distribution, we significantly enhance the generalization ability of the pruning policy via the structural risk minimization (SRM) principle. The use of SRM allows us to balance the pruning policy across layers, taking into account both the accuracy on the proxy samples and the unseen data. This leads to the optimal pruning policy, especially on challenging tasks where overfitting to the proxy samples could otherwise degrade performance. We compare our pruning method with uniform pruning policies (Uniform) and without considering the generalization ability (Auto w/o Gen). Tab. 1 shows the average pruning ratio of the pruned model and the corresponding accuracy on ScienceQA, where both automatic pruning and generalization improvement significantly contribute to the accuracy-efficiency trade-off on the acquired pruning policy. Uniform pruning policy ignores the layer-wise importance variance of pre-trained LVLMs, and searching the pruning policy on limited proxy samples without considering the generalization gap substantially degrades the performance of the policy on downstream tasks due to the overfitting.

Influence of number of proxy samples and token length: The pruning results and training cost are influenced by the number of proxy samples n and the length of tokens s during the pruning process. To find the optimal trade-off between the accuracy-efficiency trade-off and the search cost, we varied the value of n and s to observe their effects on the pruning outcomes and computational cost. Fig. 3 demonstrates the results where the pruning ratio constraint was set

Table 3. The accuracy on ScienceQA for LLaVA-SQA-7B and LLaVA-v1.5-7B, where we evaluate the models across a wide range of domains. Since LLM-Pruner fully loses its question-answering inference capacity when the pruning ratio is above 0.4, comparisons for a pruning ratio of 0.5 are not considered.

	Ratio	Method	Subject			Context Modality			Grade		Average
			NAT	SOC	LAN	TXT	IMG	NO	G1-6	G7-12	
LLaVA-SQA-7B	0.0	-	89.39	96.06	85.64	88.71	87.65	88.50	90.93	87.80	89.91
	0.3	LLM-Pruner	39.34	27.22	47.55	39.69	34.56	43.76	38.22	40.21	38.93
		SparseGPT	88.28	95.16	85.45	87.39	86.71	88.01	89.54	88.00	88.99
		Ours	89.39	93.36	86.45	88.51	86.47	89.34	89.98	88.53	89.46
	0.4	LLM-Pruner	0.18	0.11	0.64	0.20	0.15	0.49	0.18	0.46	0.28
		SparseGPT	80.73	89.65	79.82	80.40	80.17	81.32	83.99	79.43	82.36
		Ours	89.17	91.34	85.37	88.27	85.57	88.57	89.39	87.54	88.73
	0.5	SparseGPT	80.28	89.54	73.45	79.23	78.98	77.84	82.56	76.66	80.45
		Ours	83.61	90.44	75.91	82.80	83.54	79.51	84.73	80.03	83.05
LLaVA-v1.5-7B	0.0	-	62.57	69.07	65.55	62.81	63.26	63.90	68.65	57.61	64.70
	0.3	LLM-Pruner	21.67	11.36	12.73	23.17	21.52	13.45	16.30	18.79	17.19
		SparseGPT	58.88	68.05	63.45	59.24	61.87	61.39	65.12	56.36	61.99
		Ours	64.03	67.15	69.36	63.20	62.96	68.9	68.98	60.84	66.07
	0.4	LLM-Pruner	0.18	0.11	0.45	0.20	0.15	0.35	0.18	0.33	0.24
		SparseGPT	56.97	65.80	60.55	57.04	59.44	58.40	63.44	63.13	59.75
		Ours	60.35	67.27	66.91	60.26	61.77	64.53	66.74	57.68	63.50
	0.5	SparseGPT	53.51	57.71	56.82	54.01	54.64	54.98	56.86	52.34	55.25
		Ours	56.57	56.13	60.55	56.79	54.24	58.54	59.73	53.53	57.51

to 0.4. Increasing the number of proxy samples and the token lengths can both improve the accuracy given the resource budget, while also brings significant search cost burdens to the automatic pruning framework. Meanwhile, the marginal improvement becomes negligible when the number of proxy numbers and the token length exceed 64 and 256 respectively. In order to acquire the satisfying pruning policy with acceptable search cost, we select the number of proxy numbers and the token length to be 64 and 256 for the rest of the experiments. Moreover, we can also optimize the balance between resource consumption and performance, allowing them to meet their specific requirements.

Effects of search space evolution: Optimizing the projector weight in LVLMs can further significantly enhance the search space for pruning ratio, where the possible highest accuracy and generalization ability can be achieved. To verify the effectiveness of search space evolution, we conduct the ablation study to evaluate different variants: (a) the method without space evolution (No Evo), (b) the method with space evolution while failing to consider the upper bound in evolution (Evo w/o UB), which means all c_m are set to 1, (c) our method. Tab. 2 illustrates the performance including the average pruning ratio and the model accuracy. Evolving the search space can enhance the accuracy-efficiency trade-off of the acquired pruning policy, because possible highest accuracy and generalization ability are enhanced. Evaluating the upper bound of the fitness function with the average

value degrades the accuracy-efficiency trade-off because of the discrepancy of the optimization objective.

4.3. Comparison with State-of-the-art Methods

In this section, we compare the model pruning techniques in EfficientLLaVA with state-of-the-art pruning strategy. As far as we know, we are the first multimodal LVLMs pruning method, and we utilize the strategy designed for LLM pruning including LLM-Pruner [25] and SparseGPT [7] as the baseline method for further comparison. Finally, we provide two visual answers generated by LLaVA-SQA-7B.

Performance Analysis: The task of real-world scientific question answering can be leveraged as an effective way to evaluate the performance of large multimodal models. In order to demonstrate our pruning policy applicable to different models, Tab. 3 illustrates the accuracy on ScienceQA across LLaVA-SQA-7B and LLaVA-v1.5-7B architectures with variable average pruning ratios. LLaVA-SQA-7B first generates a reasoning process based on the given question, and subsequently derives the final answer from this reasoning process. On the contrary, LLaVA-v1.5-7B directly generates the final answer without an intermediate reasoning process. Given that LLM-Pruner completely loses its question-answering inference ability at a pruning ratio above 0.4, we do not make comparisons at a pruning ratio of 0.5. SparseGPT [7] can reach 50% unstructured sparsity while the accuracy degradation is 9.5%. However, it ignores the significance of various layers and generalization gap between

Table 4. The accuracy on VizWiz, MM-Vet and LLaVA-Bench for LLaVA-v1.5-7B, where we evaluate the models across a wide range of domains.

Method	Vizwiz				MM-Vet				LLaVA-Bench			
	0	0.3	0.4	0.5	0	0.3	0.4	0.5	0	0.3	0.4	0.5
LLM-Pruner		20.85	7.30	0.00		14.30	0.10	0.00		37.70	29.30	22.70
SparseGPT	50.05	46.43	41.90	38.61	35.40	21.20	20.30	17.90	55.90	57.90	52.50	48.80
Ours		52.70	52.19	50.28		33.80	33.00	31.70		58.40	54.50	53.00

the provided proxy samples and the real training data distribution, which leading the performance degradation. On the contrary, by applying the structural risk minimization principle, EfficientLLaVA formulates the generalization gap of the pruning policy and refines pruning policy candidates to enhance task performance and generalization ability. Generally speaking, with an average pruning rate of 0.5, the presented search space evolution for pruning boost the accuracy by 2.6% (80.45% vs. 83.05%) and speed up $\times 1.8$ compared by the none pruned model which get better result.

In order to demonstrate the generalization ability of our pruning policy, we also evaluate LLaVA-v1.5-7B across different dataset including ScienceQA, Vizwiz, MM-Vet and LLaVA-Bench. Compared with SparseGPT, with an average pruning rate of 0.5, EfficientLLaVA still achieves 57.51% accuracy on ScienceQA, which proves the versatility of our method, achieving excellent performance. Moreover, in complex multimodal tasks, EfficientLLaVA demonstrates superior generalization capabilities and adaptability, including Vizwiz (38.61% vs 50.28%), MM-Vet (17.90% vs 31.70%) and LLaVA-Bench (48.80% vs 53.00%) which pruning ratio is 0.5, underscoring its significant potential not only in content generation but also across a wide range of applications that require robust, context-aware processing. For pruned model with low pruning ratio outperforming dense model, we believe that removing ambiguous neurons during the pruning process can lead to better performance on some simple question-answering datasets.

Finally, We investigate acceleration of unstructured sparsity for CPU inference based on DeepSparse. Fig. 1 shows the detail of acceleration which can speed up $\times 1.5$ for LLaVA-v1.5-7B while the accuracy degradation is only 1.2%. Although LLM-Pruner inference without depending on particular kernels, it is unacceptable of the decrease of performance. Compared with SparseGPT, EfficientLLaVA achieve higher performance with the similar speed up. The observed speedups are nearly at the theoretical optimum, which suggests that unstructured sparsity acceleration for LLM inference on CPUs is a practical approach.

Visualization reasoning examples: We present several qualitative visual reasoning examples of the LLaVA-SQA-7B model using the ScienceQA dataset in Appendeix ???. Compared to LLM-Pruner, EfficientLLaVA obtains more accurate and specific answers to vision-language question pairs, which reveals the higher model capacity even with the

similar sparsity. Although LLM-Pruner can produce accurate results in the first case, it also provide irrelevant wrong information. For SparseGPT, the generated table provides incorrect answers. EfficientLLaVA can infer correct results and additionally analyzing relevant knowledge. In the second example, LLM-pruner hallucinates that melted marshmallow exists in the image leading to incorrect result. Conversely, SparseGPT accurately identifies the glass but made a critical error by overlooking the chocolate milkshake. EfficientLLaVA demonstrates the ability to comprehensively analyze the attributes of all items presented in the task. By leveraging its multimodal reasoning capabilities, it can accurately interpret both visual and textual information to identify the relevant features of each object.

5. Conclusion

In this paper, we present EfficientLLaVA, an automatically pruning method for LVLMS with limited sampled data, to achieve low latency and unaffected task performance. We institute the generalization gap of the pruning policy between the calibration samples and the unknown training set for the pruning policy using the structural risk minimization principle. By estimating via the Euclidean distance of the candidate policy set, we enhance the upper bound of the generalization ability for all policies in the pruning space. The pruning policy candidates evolve with the goal of optimizing task performance and generalization by refining the vision projector. EfficientLLaVA focuses on minimizing the generalization gap, allowing it to generalize across various scenarios after a single search. Extensive experiments on question-answer datasets demonstrate that EfficientLLaVA achieves competitive performance compared to SOTA pruned LVLMS.

While EfficientLLaVA has significantly improved generalization ability and maintained the performance of pruned LVLMS, there are still areas for future work. We aim to develop more efficient search algorithms to reduce the search cost and design a pruning policy that reduces the activation matrix to achieve higher compression ratios.

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